

Latent Representation Probing for Toxic Comment Detection

Team Mountain Zebra (TRI AI GOOGLE COHORT 10)

01 Why Toxicity Detection Needs Context

The Scale Problem

- Millions of comments daily make manual moderation impossible.
- Standard keyword filters miss subtle semantic context and sarcasm.
- Fine-tuned classifiers require significant compute for deployment.

The Efficiency Gap

- Full LLM inference is computationally prohibitive at scale.
- High latency creates bottlenecks in real-time user interactions.
- Need for lightweight, contextual, and accurate detection.

02 Technical Approach: Probe the Latent Space



We extract internal activation vectors from specific transformer layers of a frozen base model (e.g., Gemma). A lightweight probe is trained to distinguish toxic from benign text based on these high-level semantic features.

03 Dataset and Feature Extraction

Source & Splits

- Public Toxicity Benchmarks (Civil Comments / Jigsaw)
- Train / Test Split: 80% / 20% (800 / 200 samples)
- Binary Labels: 0 (Benign) vs 1 (Toxic)

Extraction Pipeline

- Base Model: google/gemma-2b
- Layer Selection: Layer 12 (intermediate transformer layer)
- Pooling: Last-token activation extraction
- Library: TransformerLens / Hugging Face

04

Evaluation Framework

Experimental Results

Metric	Baseline (Keyword / TF-IDF)	Probing Classifier (Gemma + Logistic Regression)
Accuracy	0.68 (68.0%)	0.91 (91.2%)
Precision	0.62 (62.5%)	0.89 (89.4%)
Recall	0.55 (55.0%)	0.87 (87.1%)
F1-Score	0.58 (58.3%)	0.88 (88.2%)
ROC-AUC	0.65 (0.650)	0.95 (0.954)

Supplied benchmark values compare a keyword / TF-IDF baseline with the latent probing classifier. The evaluation focuses on the performance gain of probing over traditional rule-based features.

05 Impact, Risk, and Responsible Use

Safety Impact

Community Safety: Real-time detection to filter harmful content.

Moderator Wellbeing: Reducing mental burden via automated screening.

Operational Efficiency

Low Latency: Target < 50ms classification per post.

Energy Savings: Avoiding full LLM fine-tuning and inference.

Risk Mitigation

False Positives: Calibrated thresholds and human-in-the-loop review.

Bias: Continuous evaluation on adversarial and OOD test sets.

Conclusion

Latent representation probing offers a lightweight, contextual, and efficient alternative for toxic comment detection at scale.

Next: Multi-label Toxicity & Cross-lingual Probing